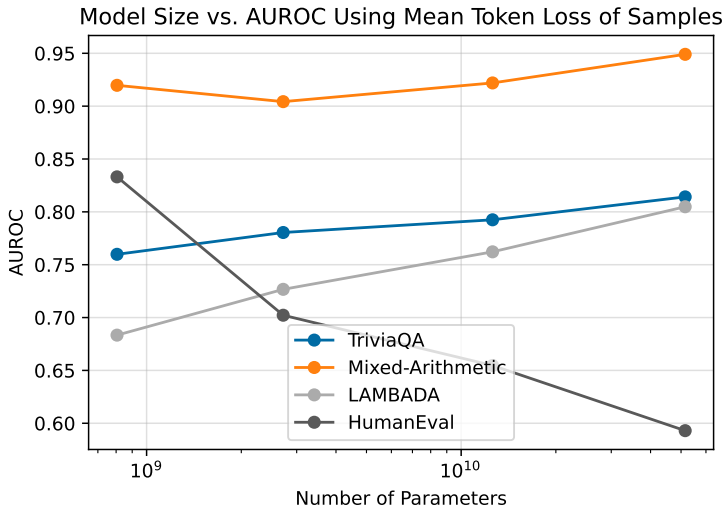

Figure 27 Model size vs. AUROC trends for several evals. ....

to evaluate model responses. Our final dataset consisted of 8000 Python functions and their corresponding tests, which was then separated out into a 6000/2000 train/test split.

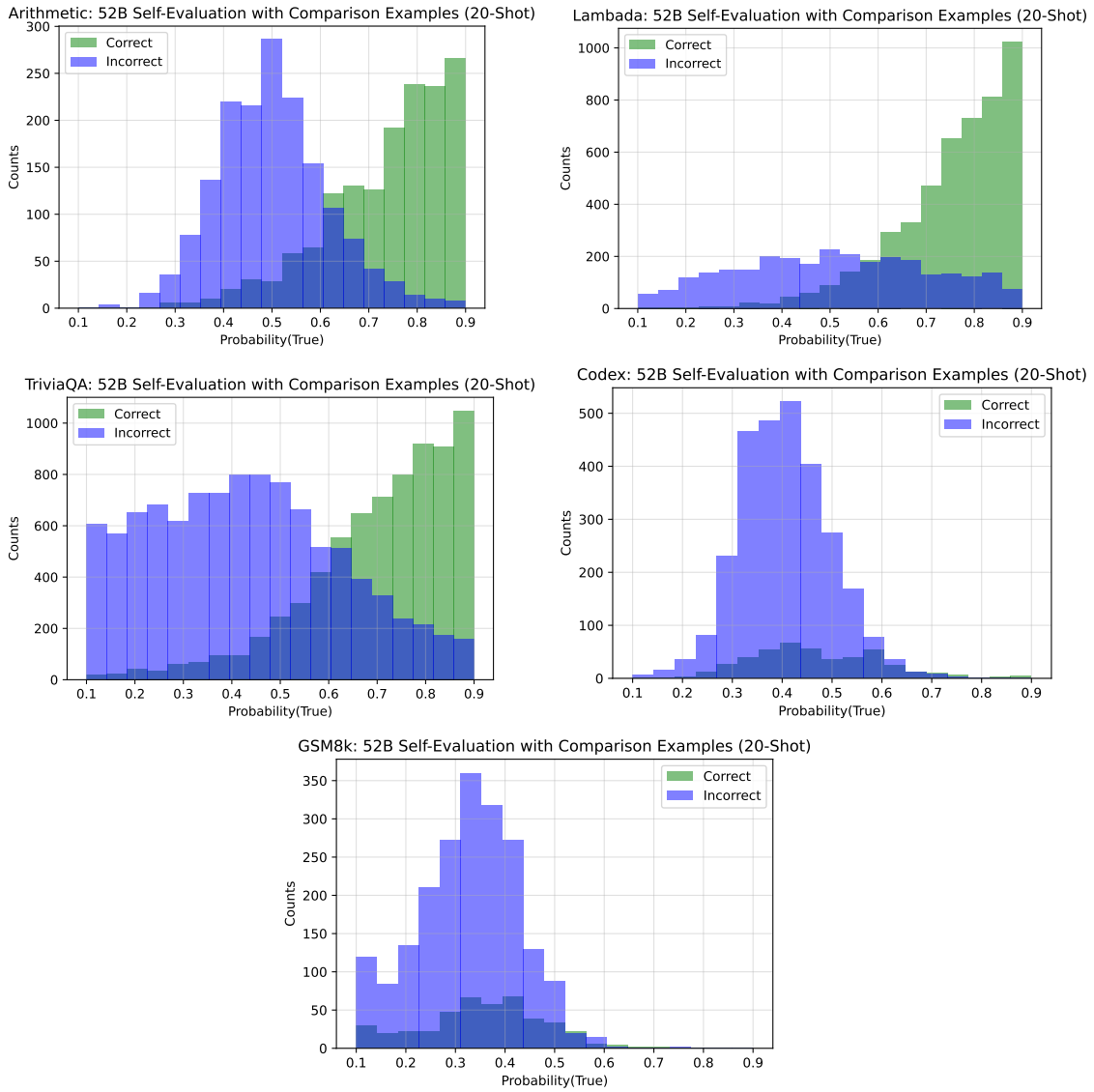

Figure 28 Models self-evaluate their own samples by producing a probability P(True) that the samples are in fact correct. Here we show histograms of P(True) for the correct and incorrect samples for each evaluation, in the evaluation paradigm where models also get to see a total of five $T = 1$ samples for the same question, in order to improve their judgment.

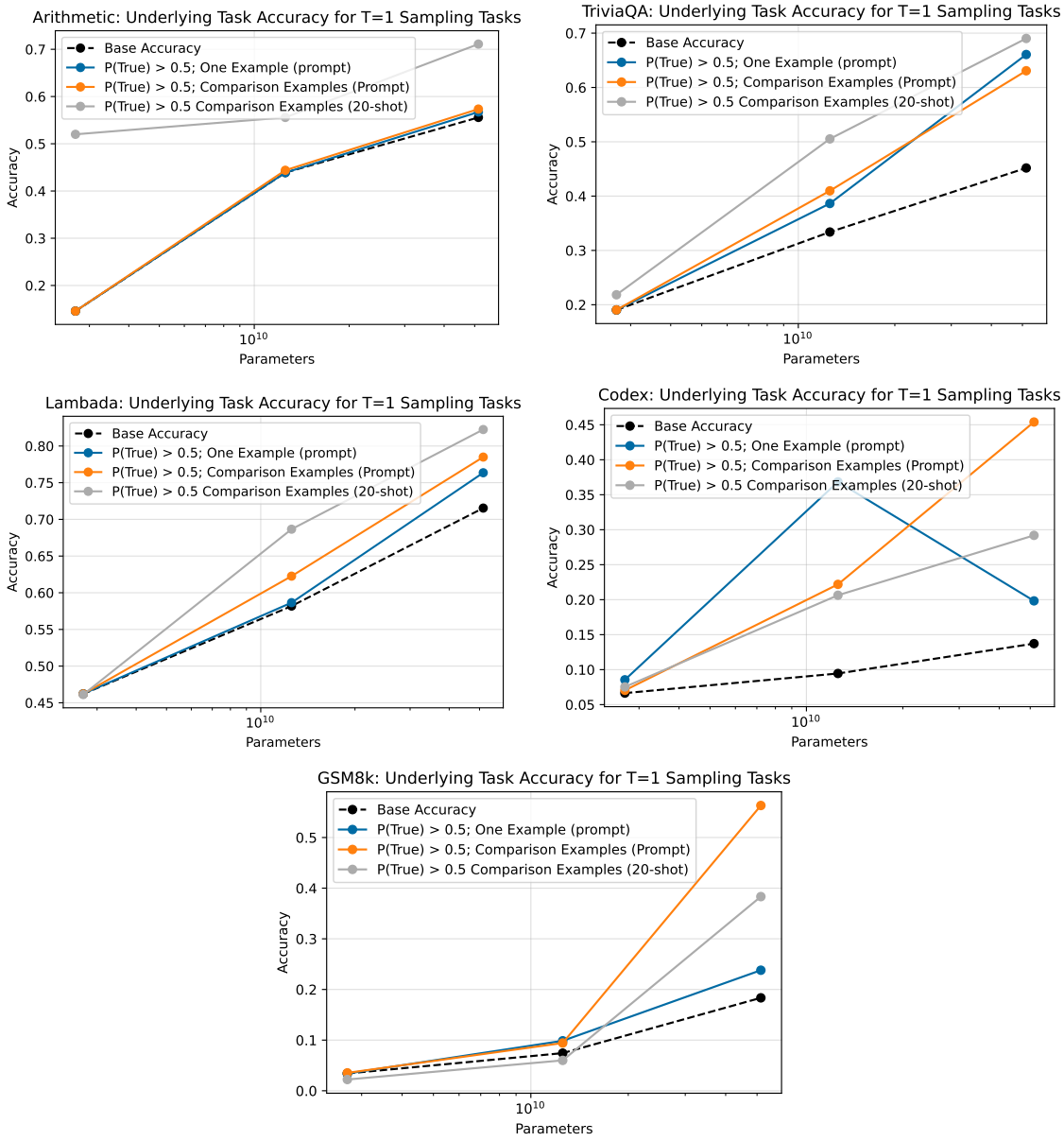

Figure 29 We show the base accuracy of our models on various sampling tasks, and then the accuracy among the responses where via self-evaluation we have P(True) > 0.5. For P(True) we evaluate both 20-shot and 0-shot with a prompt; in both cases we provide the model with five comparison examples to consider before it makes its judgment. Evaluating few-shot is crucial for obtaining good calibration for P(True), though it appears that the prompt actually performs better than few-shot for codex and GSM8k, possibly because these have long-form answers.